CS & IT ENGINEERING

DIGITAL LOGIC

Minimization



Lecture No. 1



By - CHANDAN SIR

Recap of Previous Lecture



Logic Gates questions practice

Topics to be Covered



BOOLEAN ALGEBRA

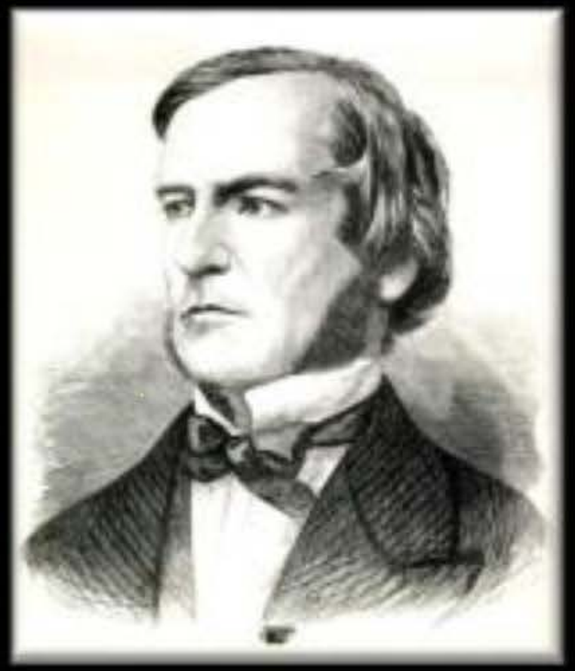
- D-MORGAN'S LAW
- DUAL & SELF DUAL
- QUESTIONS PRACTICE
- DISCUSSION



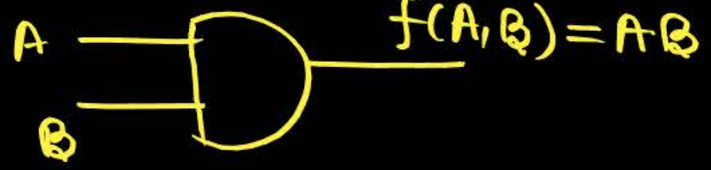
Topic : Laws of Boolean Algebra

- **1854- George Boole**

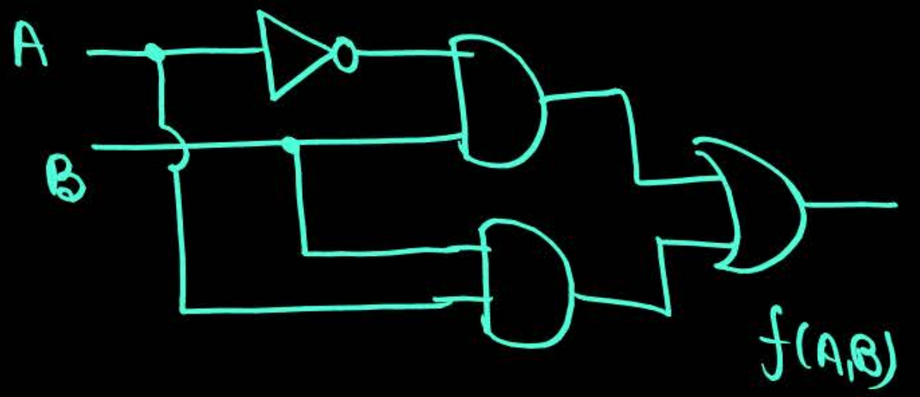
"An Investigation of Law of Thoughts"



Ex.1 $f(A,B) = A \cdot B$



Ex.2 $f(A,B) = \bar{A}B + AB$





Topic : Laws of Boolean Algebra

Boolean Algebra

Boolean function

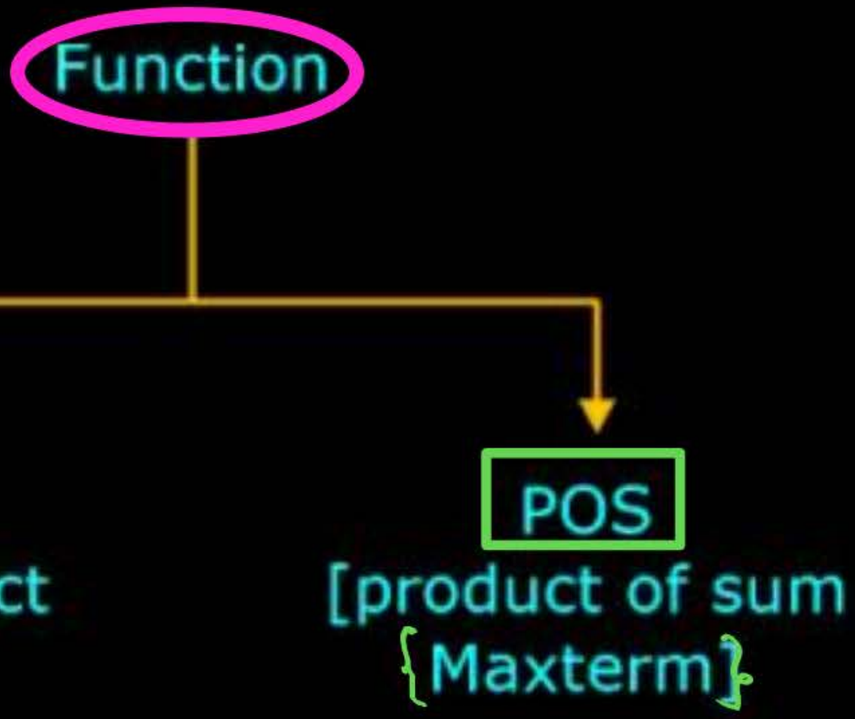
↳ Combination of inputs on which output will depends.



Topic : Laws of Boolean Algebra

Boolean Algebra

Minterm = $\overline{\text{Maxterm}}$





Topic : Laws of Boolean Algebra

Boolean Algebra

SOP

$f(A,B) = \bar{A}\bar{B} + AB$

Variable

Term Term

minterm

Literals

POS

$f(A,B) = (\bar{A}+B)(A+\bar{B})$

Term-1 Term-2

Max Term

Decimal	A B	minterm	Maxterm	function
0	0 0	$\bar{A}\bar{B}$	$A+B$	1
1	0 1	$\bar{A}B$	$A+\bar{B}$	0 ✓
2	1 0	$A\bar{B}$	$\bar{A}+B$	0
3	1 1	AB	$\bar{A}+\bar{B}$	1

Standard canonical

Standard canonical SOP form

SOP

$$f(A,B) = \bar{A}\bar{B}\cdot 1 + \bar{A}B\cdot 0 + A\bar{B}\cdot 0 + AB\cdot 1$$

$$= \bar{A}\bar{B} + AB$$

$$= m_0 + m_3$$

$$= \Sigma m(0,3)$$

Standard canonical POS form

POS

$$F(A,B) = (A+B+1)\cdot(A+\bar{B}+0)\cdot(\bar{A}+B+0)(\bar{A}+\bar{B}+1)$$

$$= 1\cdot(A+\bar{B})\cdot(\bar{A}+B)\cdot 1$$

$$= (A+\bar{B})\cdot(\bar{A}+B)$$

$$= M_1\cdot M_2$$

$$= \Pi M(1,2)$$



Topic : Laws of Boolean Algebra

ये ऐसे ही लीखाईं Ex. के लिये

Decimal	ABC	Min Term	Max Term	Function
0	000	$\bar{A}\cdot\bar{B}\cdot\bar{C}$	$A+B+C$	1
1	001	$\bar{A}\bar{B}C$	$A+B+\bar{C}$	0
2	010	$\bar{A}B\bar{C}$	$A+\bar{B}+C$	1
3	011	$\bar{A}BC$	$A+\bar{B}+\bar{C}$	0
4	100	$A\bar{B}\bar{C}$	$\bar{A}+B+C$	0
5	101	$A\bar{B}C$	$\bar{A}+B+\bar{C}$	1
6	110	$AB\bar{C}$	$\bar{A}+\bar{B}+C$	0
7	111	ABC	$\bar{A}+\bar{B}+\bar{C}$	1



Topic : Laws of Boolean Algebra

Boolean Function : It is the combination of inputs on which output is depends.



Topic : Laws of Boolean Algebra

Theorem

1) Distribution theorem

$(A + B)(A + C)$ $\qquad A \cdot (B + C) = AB + AC$

2) Consensus theorem

$AB + \overline{A}C + BC$

3) Transpose theorem

$(A + B) + (\overline{A} + C)$

4) D-Morgan's Law

$\overline{ABC} = \overline{A} + \overline{B} + \overline{C}$

$\overline{A + B + C} = \overline{A} \cdot \overline{B} \cdot \overline{C}$



Topic : Laws of Boolean Algebra

Theorem

5) Annulment Law

$A \cdot 0 = 1, \qquad A + 1 = 1$

6) Identity Law

$A + 0 = A, \qquad A \cdot 1 = A$

7) Idempotent Law

$A + A = A, \qquad A \cdot A = A$

8) Absorptive Law

$A + AB = A$

$A \cdot (A + B) = A$

#Q. Find the minimum number of the NAND gate required to implement the Boolean function given below:

$f(A, B, C) = A + ABC + AB\overline{C}$

#Q. Minimize the expression:

$f(A, B) = A + A\overline{B}$

#Q. Minimize the expression.

$f(A, B) = \overline{A}\,\overline{B} + \overline{A}B + AB$

#Q. Minimize the expression.

$f(A, B) = \overline{A}\,\overline{B} + \overline{A}B + A\overline{B} + AB$

#Q. Minimize the expression.

$f(A, B) = \overline{A}B + A\overline{B}$

#Q. Minimize the expression.

$f(A, B) = AB + \overline{A}C + BC$

#Q. Minimize the expression.

$f(A, B, C) = \overline{A}\,\overline{B} + \overline{A}C + \overline{B}\,\overline{C}$

#Q. Minimize the expression.

$f(A, B, C) = (A + B)(A + C)(\overline{B} + C)$

#Q. Write the function for truth table and minimize it.

A	B	Y(O/p)
0	0	1
0	1	0
1	0	1
1	1	1

#Q. Write the function for truth table and minimize it.

A	B	Y(O/p)
0	0	C
0	1	$\overline{C}$
1	0	1
1	1	1

#Q. Two way switch is an example of which logic?

A AND

B OR

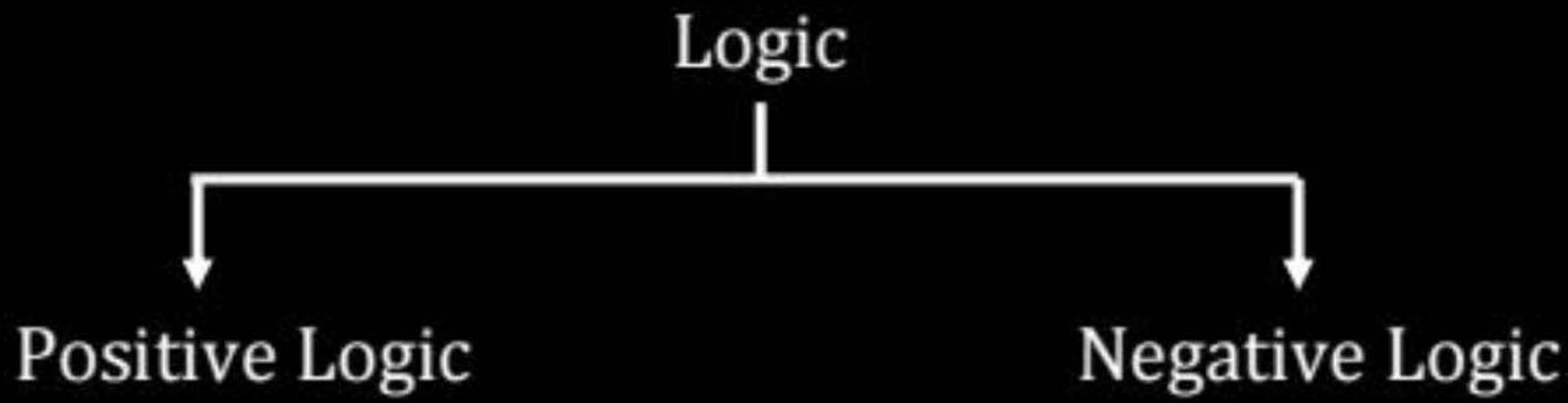
C X-OR

D NAND

#Q. If we have ‘4’ variable, then total different expression will be?



Dual





Topic : Laws of Boolean Algebra

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

A	B	Y
1	1	1
1	0	1
0	1	1
0	0	0



Topic : Laws of Boolean Algebra

A	B	Y
0	0	
0	1	
1	0	
1	1	

A	B	Y

Dual is a process in which the positive logic converted into negative logic and vice versa



Dual

i) Positive logic $\xrightarrow{\text{Dual}}$ negative logic

ii) NPN trans $\rightarrow$ PNP trans

iii) N-MOS $\rightarrow$ P-MOS

iv) $0 \leftrightarrow 1$

v) $\cdot \leftrightarrow +$



Topic : Laws of Boolean Algebra

Dual

vi) AND ↔ OR

vii) NAND ↔ NOR

viii) EX – OR ↔ EX – NOR

ix) BUFFER ↔ BUFFER

x) INVERTER ↔ INVERTER



Topic : Laws of Boolean Algebra

Self-Dual Functions

"An Investigation of Law of Thoughts"



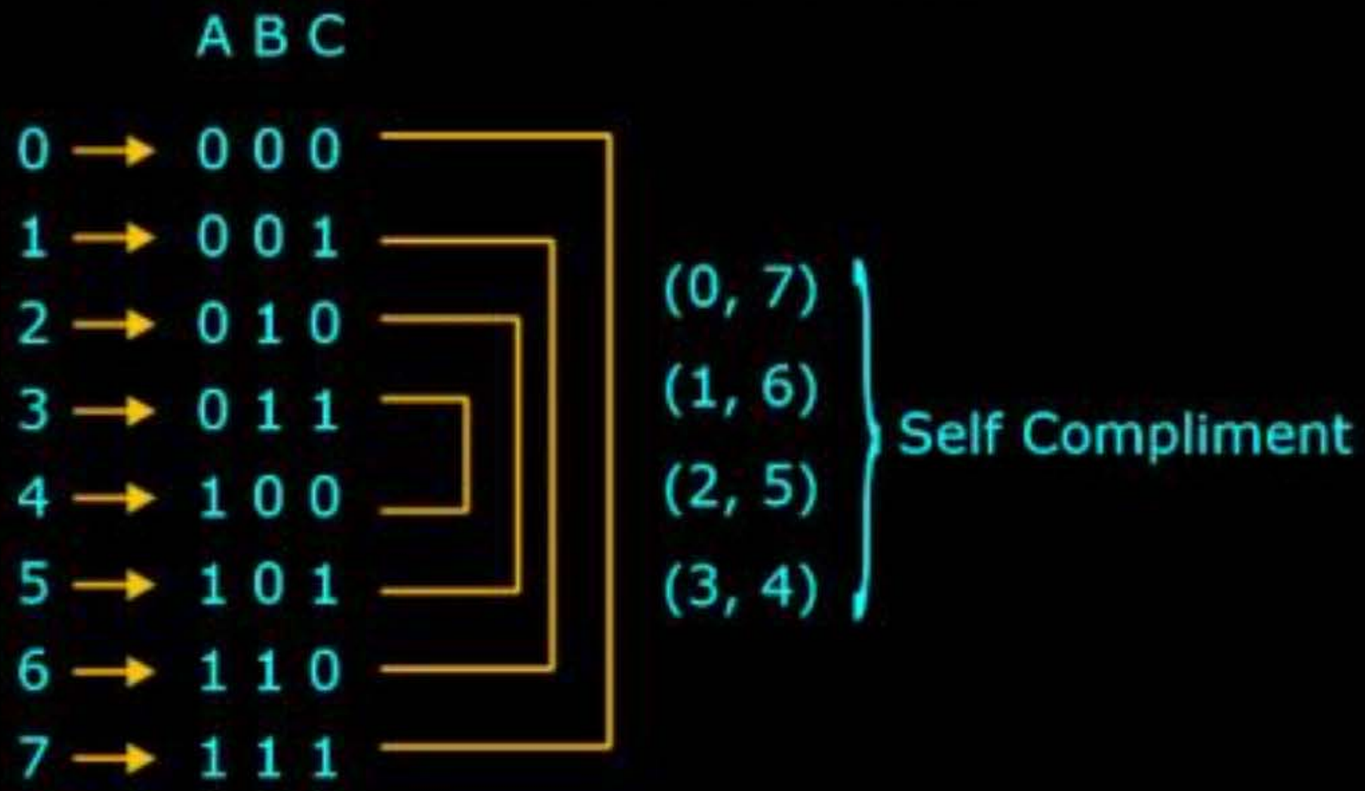
Note:

With the n variable we can form total 2^{2^n} different expression in that exp. $2^{2^{n-1}}$ expression are self dual.



Topic : Laws of Boolean Algebra

Mutual Exclusive Functions



n-variable – [Maxterm = Minterm = 2^n]



2 mins Summary

Topic Explain

- D-MORGAN'S LAW
- DUAL & SELF DUAL
- QUESTIONS PRACTICE
- DISCUSSION

Thank You GW Soldiers!

